Nilin Abrahamsen

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GitHub | Twitter | LinkedIn | Google Scholar

Experience

Voleon Capital Management – Member of Research Staff, 2024–Present

- Led development and launch of a new portfolio strategy.
- Mentored intern to create new alpha signals.

University of California, Berkeley – Postdoctoral Researcher, 2021–2024

- Developed optimization methods and neural architectures for computational physics.
- Extensive use of ML frameworks (JAX, PyTorch) and JIT-compiled Python (Numba-CUDA).

Education

Massachusetts Institute of Technology – PhD in Mathematics

- Tensor network algorithms for computational physics (Advisors: Peter Shor and Jon Kelner).

University of Copenhagen – BS and MS in Mathematics

- Awarded Best MS Thesis in Mathematics in Denmark (Danish Mathematical Association).

Projects

NanoGPT speedrun, optimization track – May 2026

- Broke the overall record 3 times for the optimization track of the NanoGPT speedrun.
- Introduced 2 new innovations (contra-muon and radial brake) which withstood 10,000 ablation runs by PrimeIntellect.

PROMA: Projected Microbatch Accumulation for Reference-Free Proximal Policy Updates – Jan 2026 (github, arxiv:2601.10498)

- Proximal policy optimization without a reference policy.
- Projects away high-variance subspace of the policy gradient to approximate the natural gradient.
- Implemented in VeRL.

Skills & Languages

Programming & Tools: PyTorch, JAX, Numba-CUDA, Reinforcement Learning, Context Engineering
Languages: English, Danish, Mandarin Chinese (professional working proficiency)